

Project Initialization and Planning Phase

Date	27 June 2026
Team ID	-
Project Title	Smart Lender
Maximum Marks	5 Marks

PROJECT PROPOSAL (PROPOSED SOLUTION) REPORT

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

1. Project Overview	
Objective	Develop a machine learning-based web application that predicts the creditworthiness of loan applicants, enabling banks and financial institutions to make faster, more accurate, and data-driven loan approval decisions. The system aims to reduce manual effort, minimize loan default risk, and improve operational efficiency.
Scope	The Smart Lender project focuses on automating the loan approval process using machine learning algorithms. It analyzes applicant information such as income, education, employment status, credit history, loan amount, and property area to predict loan approval. The application is developed using Python and Flask and is designed for deployment on IBM Cloud.
2. Problem Statement	
Description	Traditional loan approval processes rely heavily on manual verification, making them time-consuming, inconsistent, and prone to human error. Financial institutions often face challenges in accurately assessing applicant creditworthiness, resulting in delayed decisions and increased loan default risks.

Impact	Implementing an intelligent loan approval prediction system will: • Reduce loan processing time. • Improve decision accuracy. • Minimize non-performing assets (NPAs). • Enhance customer satisfaction. • Increase operational efficiency for banks and financial institutions.
3. Proposed Solution	
Approach	The proposed solution uses supervised machine learning algorithms to analyze applicant data and predict loan approval. Multiple classification models—Decision Tree, Random Forest, K-Nearest Neighbors (KNN), and XGBoost—are trained and evaluated. The best-performing model (XGBoost) is integrated into a Flask web application to provide real-time predictions.
Key Features	• Machine learning-based loan approval prediction. • Real-time applicant evaluation. • Fast and accurate decision-making. • User-friendly web interface developed using Flask. • Supports multiple classification algorithms. • Reduces manual verification effort. • Helps identify high-risk loan applicants.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware	Computing Resources	Intel Core i5 / Apple M1 Processor
	Memory	8 GB RAM
	Storage	256 GB SSD

Software	Frameworks	Flask
	Libraries	scikit-learn, pandas, numpy, matplotlib, seaborn, xgboost
	Development Environment	Visual Studio Code, Jupyter Notebook
Data	Data Source, Size, Format	Kaggle Loan Prediction Dataset, 614 records, CSV